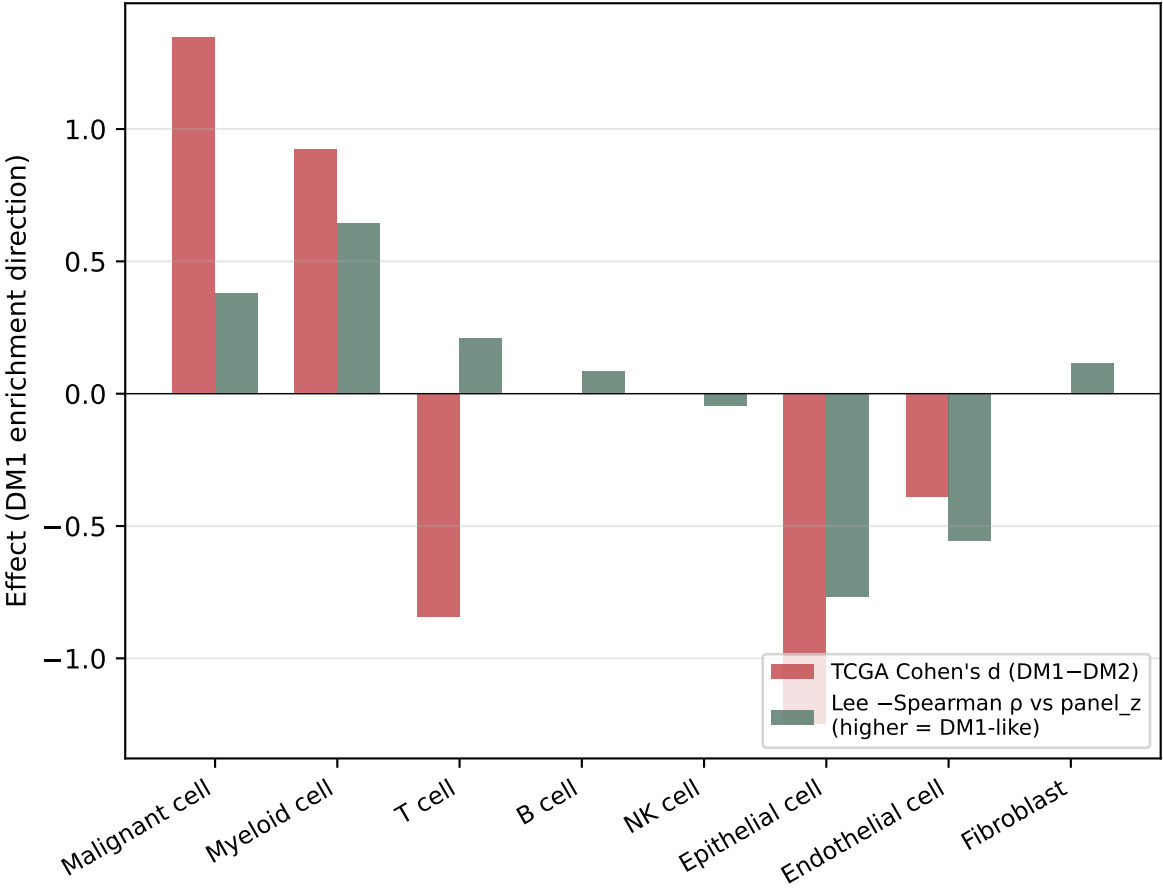
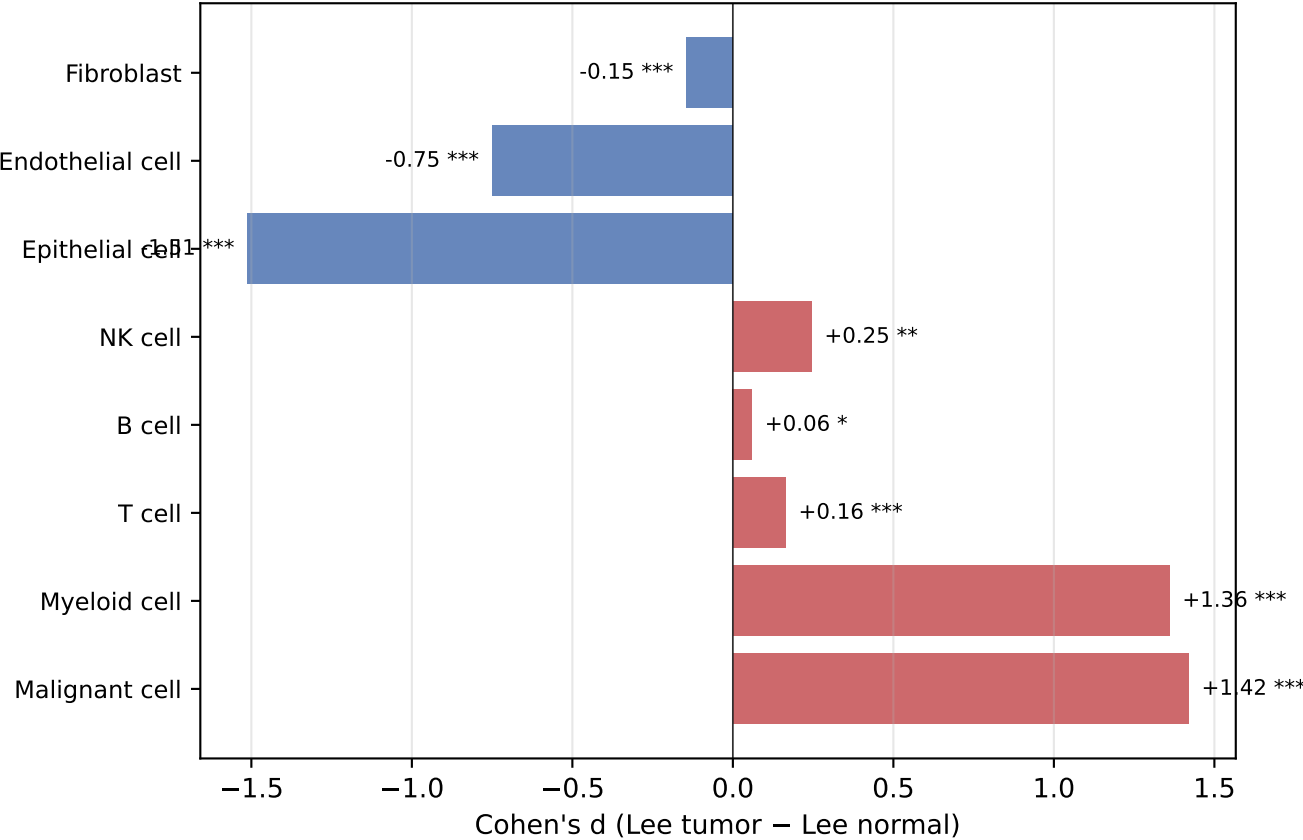


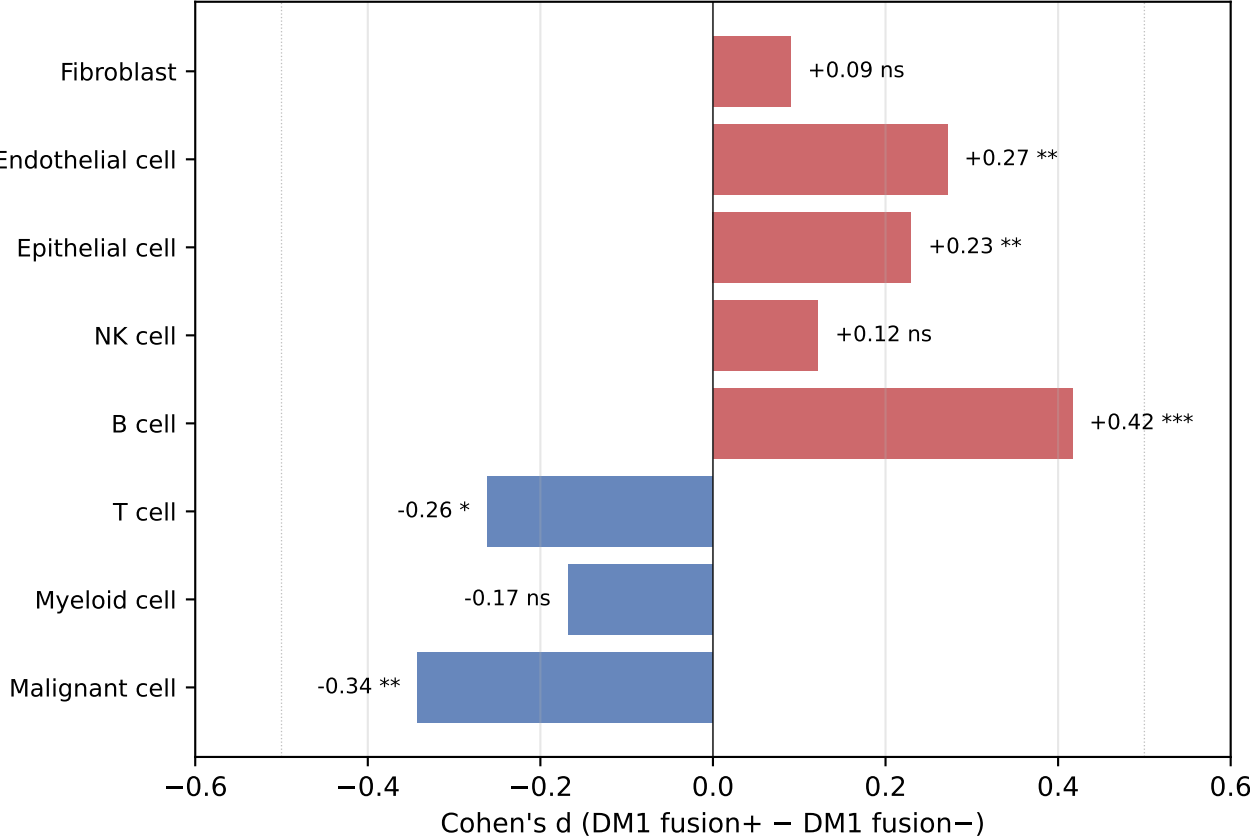
(A) Cross-cohort direction consistency: TCGA nu-SVR vs Lee/GSE213647
(both: positive = DM1-like / less-differentiated direction)



(B) Lee tumor vs normal cell-type fractions (n=369 vs 263)



(C) Within-DM1: fusion+ vs fusion- cell-type fractions
(kinase fusion: RET/NTRK/ALK/BRAF/PAX8/PPARG; n=74 vs 391)



v3 EXTENSIONS – KEY FINDINGS

(A) CROSS-COHORT DIRECTION CONSISTENCY
TCGA Cohen's d (DM1-DM2, nu-SVR) and Lee -Spearman ρ vs panel_z (tumor-only n=369) agree on direction for 7/8 cell types:
Same direction: Malignant↑, Myeloid↑, Epithelial↓, Endothelial↓, Fibroblast~
Discordant: T cell – TCGA d=-0.84 vs Lee -ρ=+0.21 (likely tumor-purity confound dominates TCGA; Lee's continuous gradient resolves nuance)

(B) LEE TUMOR vs NORMAL
Tumor enriched: Malignant (d=+1.42), Myeloid (+1.36), T (+0.16), NK
Tumor depleted: Epithelial (d=-1.51, normal-thyroid epithelium), Endothelial (-0.75)
Direction-consistent with TCGA DM1 vs DM2 (DM1 ≈ "tumor-like").

(C) WITHIN-DM1 FUSION+ vs FUSION-
All |Cohen's d| ≤ 0.42 (max B cell +0.42, then Malignant -0.34).
Most cell types show negligible composition difference between fusion-positive and fusion-negative DM1 tumors.

INTERPRETATION

The within-DM1 fusion+ vs fusion- composition equivalence (panel C, all |d|≤0.42) supports the manuscript's §2.4a / Fig 8 claim that the epigenetic silencing program is fusion-independent: if the two sub-populations had grossly different cell-type compositions, the methylation signal could be confounded; instead they are near-identical in composition, consistent with a shared epigenetic state across fusion strata.

CAVEATS

- Lee panel_z is continuous, not dm_like binary; cross-cohort alignment uses sign-flip convention (negative ρ = DM1-direction).
- T-cell discrepancy between TCGA Cohen's d and Lee correlation is unresolved; likely reflects purity confound differing between binary and continuous score frames.
- cbio_sv kinase-fusion definition restricted to events where eventInfo contains RET/NTRK/ALK/BRAF/PAX8/PPARG; alternate fusion calls in MAF/STAR-Fusion may capture additional cases.